

ChatGPT Reviews Analysis – Complete Video Presentation Script

Introduction

Hello everyone.

Today, I will be presenting my project on ChatGPT Reviews Analysis using Python. The goal of this project is to analyze customer reviews of ChatGPT and understand how users feel about the product by applying Natural Language Processing and Sentiment Analysis techniques.

Customer reviews contain valuable information about user experiences, satisfaction levels, and areas where improvements may be needed. Since manually reading thousands of reviews is time-consuming, data analytics and NLP help us automatically extract meaningful insights from large amounts of textual feedback.

Through this project, I aim to answer the following business questions:

1. What overall sentiment do users express about ChatGPT?
2. How strongly do users feel about their opinions?
3. What features do users appreciate the most?
4. What aspects receive criticism?
5. How well do user ratings align with sentiment scores?

By answering these questions, we can provide actionable insights that help improve customer satisfaction and product quality.

Dataset Understanding

Let me start by understanding the dataset.

The dataset contains customer reviews collected from users of ChatGPT. The main fields include review text, ratings, dates, and other metadata related to user feedback.

The first step is loading the dataset into Python using Pandas.

After loading the dataset, I inspect the first few rows using the `head()` function to understand its structure and verify that the data has loaded correctly.

Next, I use `info()` and `describe()` functions to understand data types, identify missing values, and generate summary statistics.

Understanding the dataset before analysis is extremely important because it helps identify data quality issues and guides the preprocessing strategy.

Data Cleaning and Preparation

Data cleaning is one of the most important stages of any data analysis project.

Poor-quality data can lead to misleading results, so I perform several preprocessing steps before beginning the analysis.

First, I standardize column names by converting them to lowercase and removing unnecessary spaces. This improves consistency and makes coding easier.

Next, I check for missing values across all columns.

For missing review texts, I replace null values with blank strings because sentiment analysis requires text inputs and missing values could otherwise generate errors.

I then ensure that ratings are stored as numeric values rather than text.

Similarly, date columns are converted into proper datetime format to support time-based analysis.

These preprocessing steps ensure that the dataset is clean, structured, and ready for further analysis.

Exploratory Data Analysis

Once the data is prepared, I perform Exploratory Data Analysis, commonly known as EDA.

EDA helps uncover patterns, trends, and anomalies before applying advanced analytical techniques.

Rating Distribution

The first visualization shows the distribution of ratings.

This chart helps us understand how users generally rate ChatGPT.

From the chart, we can observe that higher ratings appear more frequently than lower ratings.

This indicates that the majority of users have a positive experience with ChatGPT.

Review Length Analysis

Next, I calculate review length by counting the number of characters in each review.

Analyzing review lengths helps identify how detailed user feedback tends to be.

The histogram shows that while many reviews are short, some users provide highly detailed feedback describing their experiences.

Longer reviews often contain richer sentiment information and more specific feature feedback.

Rating Trends Over Time

I also analyze how ratings change over time.

This visualization allows us to identify whether customer satisfaction is improving, declining, or remaining stable.

Monitoring trends over time is valuable because it helps organizations evaluate the impact of product updates and improvements.

Through these EDA techniques, we gain a better understanding of the overall structure and behavior of the data before applying sentiment analysis.

Sentiment Analysis

Now I move to the core part of the project: Sentiment Analysis.

Sentiment Analysis is a Natural Language Processing technique used to determine the emotional tone behind textual data.

For this project, I use the TextBlob library.

TextBlob provides two important sentiment metrics:

1. Polarity
2. Subjectivity

Polarity

Polarity measures whether a review is positive, neutral, or negative.

The polarity score ranges from -1 to +1.

A score close to -1 indicates negative sentiment.

A score near 0 indicates neutral sentiment.

A score close to +1 indicates positive sentiment.

Using TextBlob, I calculate polarity scores for every review.

Subjectivity

Subjectivity measures how opinion-based a review is.

Its values range from 0 to 1.

A score closer to 0 means the review is objective and fact-based.

A score closer to 1 means the review is highly opinionated and personal.

Subjectivity helps us understand how emotionally driven user feedback is.

Sentiment Classification

After calculating polarity scores, I classify each review into one of three categories:

Positive

Neutral

Negative

If polarity is greater than zero, the review is labeled Positive.

If polarity is less than zero, the review is labeled Negative.

Otherwise, it is labeled Neutral.

This classification simplifies interpretation and allows easier visualization of overall sentiment trends.

Sentiment Distribution Analysis

The sentiment distribution chart summarizes the number of positive, neutral, and negative reviews.

The results clearly show that positive reviews dominate the dataset.

This indicates that most users are satisfied with ChatGPT and generally share favorable experiences.

Neutral reviews represent balanced opinions without strong emotional expression.

Negative reviews exist but constitute a smaller portion of total feedback.

This overall distribution suggests strong customer satisfaction and positive market perception.

Subjectivity Analysis

Next, I analyze subjectivity scores.

The histogram of subjectivity helps us understand how users express their opinions.

Most reviews fall within the moderate-to-high subjectivity range.

This indicates that users frequently share personal experiences and opinions when reviewing ChatGPT.

This is expected because product reviews often contain emotional reactions, preferences, and personal evaluations.

Understanding subjectivity provides additional context beyond simple positive or negative sentiment classification.

Sentiment versus Ratings

One important validation step is comparing sentiment scores with user ratings.

If sentiment analysis is working correctly, reviews with higher ratings should generally exhibit higher sentiment polarity.

To verify this relationship, I create a visualization comparing ratings and polarity scores.

The results show a strong positive relationship.

Higher-rated reviews typically have higher sentiment scores, while lower-rated reviews tend to exhibit negative sentiment.

This alignment validates the reliability of the sentiment analysis approach used in the project.

Positive Review Analysis

After understanding overall sentiment patterns, I focus specifically on positive reviews.

I filter reviews with ratings of four stars and above.

These reviews help identify the strengths and features users appreciate most.

To prepare the text for keyword analysis, I perform text cleaning by:

Removing punctuation

Removing special characters

Converting text to lowercase

Removing unnecessary symbols

Once cleaned, I combine all positive reviews into a single text corpus.

Keyword Extraction

Next, I calculate word frequencies to identify commonly mentioned positive terms.

The analysis reveals several frequently occurring words, including:

Helpful

Useful

Accurate

Fast

Easy

Productive

Assistant

Information

Answers

Learning

These keywords suggest that users highly value ChatGPT for its helpfulness, speed, ease of use, and ability to provide accurate information.

This analysis helps identify the features that contribute most strongly to customer satisfaction.

Word Cloud Visualization

To provide a visual representation of keyword frequencies, I generate a Word Cloud.

In a Word Cloud, larger words represent higher frequencies.

The visualization highlights the most common positive terms appearing in user reviews.

Word Clouds provide an intuitive and engaging way to summarize large amounts of textual data.

They help stakeholders quickly understand the dominant themes in customer feedback.

Key Insights

Based on the analysis, several important insights emerge.

Insight One:

The majority of reviews express positive sentiment, indicating strong overall user satisfaction.

Insight Two:

Sentiment polarity aligns closely with user ratings, validating the effectiveness of the sentiment analysis model.

Insight Three:

Users consistently praise ChatGPT for being helpful, accurate, easy to use, and time-saving.

Insight Four:

Positive reviews frequently emphasize productivity improvements and learning support.

Insight Five:

Negative reviews primarily mention occasional inaccuracies, response limitations, and subscription-related concerns.

These findings provide a balanced understanding of both strengths and improvement opportunities.

Recommendations

Based on the insights obtained from this analysis, I propose the following recommendations.

First, continue enhancing features that users already value, particularly accuracy, speed, and ease of use.

Second, monitor recurring complaints found in negative reviews and prioritize addressing common concerns.

Third, conduct periodic sentiment monitoring to track changes in customer perception over time.

Fourth, use review analytics as part of an ongoing customer feedback strategy to identify emerging issues early.

Implementing these recommendations can help improve user satisfaction and strengthen product performance.

Conclusion

To conclude, this project successfully analyzed customer reviews of ChatGPT using Python, Natural Language Processing, and Sentiment Analysis techniques.

Starting from data cleaning and preparation, I explored the dataset, measured sentiment polarity and subjectivity, classified reviews into sentiment categories, extracted frequently mentioned keywords, and created visualizations to communicate findings effectively.

The results indicate that ChatGPT receives predominantly positive feedback from users.

Customers particularly appreciate its helpfulness, accuracy, speed, and ease of use.

At the same time, the analysis identifies areas where improvements can further enhance user experience.

This project demonstrates how data analytics and NLP can transform large volumes of unstructured text into meaningful business insights and actionable recommendations.

Thank you for watching my presentation. I hope this analysis provided valuable insights into customer perceptions of ChatGPT.